Distribution-free comparison of above-ground biomass by group

Abstract

Background: A field experiment on nitrogen fertilisation in grassland plots. Methods: Above-ground biomass (g/m^2) was compared between the fertilised and unfertilised groups with a two-sided Mann-Whitney U test (asymptotic method with continuity correction). Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a field experiment on nitrogen fertilisation in grassland plots. The primary question concerns nitrogen fertilisation and above-ground biomass (g/m^2). The accompanying dataset (data.csv) contains one row per plot-record as exported from the study database, including some duplicated exports and sentinel-coded missing values; it is described in the data dictionary.

Methods

Participants and data. The export contains 268 records. Variables are listed in the data dictionary. Group membership is recorded in `treatment`.

Data cleaning and exclusions. Exact duplicate records (identical on every column) were removed, keeping the first occurrence. The value -999 denotes a missing measurement and was treated as missing. Only plots aged 18 to 80 years inclusive were analysed. Group labels in `treatment` were entered free-text and were normalised by trimming whitespace and lower-casing before use. Records with missing biomass were excluded (complete-case analysis).

Statistical analysis. Above-ground biomass (g/m^2) was compared between the fertilised and unfertilised groups with a two-sided Mann-Whitney U test (asymptotic method with continuity correction). The U statistic is reported for the fertilised group. The rank-biserial correlation was computed as 1 - 2U/(n1 n2). Medians and the difference in medians (fertilised minus unfertilised) are reported. All analyses used the cleaned analysis sample; no random resampling was used.

Results

The fertilised group ($n = 121$, median 475.90) differed from the unfertilised group ($n = 120$, median 419.50); the difference in medians was 56.40 ($U = 9670.50$, $p < 0.001$, rank-biserial $r = -0.332$).

Discussion

The analysis followed the pre-specified plan. Limitations include reliance on database exports and complete-case handling of missing values.